

Team Formation and Initial Project Idea Selection

Course: CSE 4204 – Mobile Computing Lab
Department: Computer Science and Engineering
University: Northern University of Business and Technology Khulna

Official Team Name	CSE4204-8A-T07
Section	8A
Project Title	AI Smart Campus System
Team Leader	Md. Nazmus Shakib
Leader ID	11220320852

1. Team Member List

Serial	Name	Student ID
1	Md. Nazmus Shakib	11220320852
2	Samira Akter Mitu	11220320858
3	Tanvin Sadik Dhrubo	11220320860
4	Khan Waziur Rahman	11220320861

2. Preliminary Project Description

The proposed project is an AI-Smart Campus System designed to improve academic management, communication, and student performance monitoring within a university environment. The system will provide dedicated portals for Students, Faculty Members, and Administrators. Students will be able to monitor their CGPA, attendance, routines, and official notices, while faculty members will manage student performance records including quiz, assignments, and mid-term marks. The integrated AI system will analyze academic performance, recommend suitable courses based on previous semester results, identify academically vulnerable students, and provide instant university-related assistance through an intelligent chatbot.

3. Why AI is needed in this Project

Artificial Intelligence plays a central role in making the proposed system intelligent, proactive, and data driven.

AI will be used to:

- Analyze students' previous academic performance and recommend appropriate elective and core courses.
- Detect students who are academically at-risk using attendance, quiz scores, and mid-term performance data.
- Generate intelligent academic insights for faculty members.
- Provide an AI-powered campus assistant using Gemini API that can answer questions regarding notices, exam schedules, routines, library hours, and university FAQs.
- Through intelligent recommendation, predictive analytics, and automated assistance, AI transforms the system from a traditional campus portal into a smart academic support platform.

4. Initial Technology Stack

Category	Technology
Frontend	React.js + Tailwind CSS
Backend	Laravel
Database	MySQL (XAMPP)
AI Integration	Gemini API
Version Control	Git & GitHub

5. Team Member Responsibilities

Member Name	ID	Responsibility
Tanvin Sadik Dhrubo	11220320860	Frontend Development & UI Design
Khan Waziur Rahman	11220320861	Backend Development
Md. Nazmus Shakib	11220320852	Database Management & Testing
Samira Akter Mitu	11220320858	AI Integration

6. Similar Existing Systems / Research

During the initial research phase, several existing systems and educational platforms were studied, including:

Google Classroom Moodle LMS Canvas LMS Microsoft Teams for Education. These systems provide learning management and communication functionalities, but many lack personalized AI-based academic recommendations, automated student risk detection, and campus-specific intelligent assistance. The proposed AI-Smart Campus System aims to improve the traditional academic management workflow by integrating intelligent course recommendations, predictive academic analytics, and AI-powered institutional assistance.

7. Conclusion

This project aims to combine modern web technologies with Artificial Intelligence to build an intelligent campus management ecosystem. The system will help students manage academic information, improve course planning, and access personalized guidance. Faculty members will benefit from automated performance monitoring and risk detection tools. Through AI-driven recommendations, predictive analytics, and intelligent campus assistance, the proposed system seeks to improve academic productivity, communication efficiency, and overall student success.

Submitted To:

Md. Riaz Mahmud

Assistant Professor

Department of Computer Science and Engineering

Northern University of Business and Technology Khulna